

Dynamic Programming

Chapter 04

- Numerical techniques geared towards solving for the optimal policy / optimal value function or
- given a policy, estimate the value of it
- We assume in this class that the system model, i.e; $P(s', r | s, a)$, is known
- Recall the Bellman Equation for optimality

- $$V_*(s) = \max_a \sum_{s', r} p(s', r | s, a) * (r + \gamma V_*(s'))$$

[Bellman Equation]

- $$Q_*(s, a) = \sum_{s', r} p(s', r | s, a) \left[r + \gamma \max_{a'} Q_*(s', a') \right]$$

[Q- Bellman Equation]

We begin with the Policy Evaluation or prediction procedure.

Given a policy π ,

$V_{\pi}(s) \equiv$ value of policy π in state s

We know that

$$V_{\pi}(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r|s, a) * [r + \gamma V_{\pi}(s')]$$

Simple Analysis

Suppose

$V_{\pi} =$

$$\begin{bmatrix} V_{\pi}(1) \\ V_{\pi}(2) \\ \vdots \\ V_{\pi}(n) \end{bmatrix}$$

where

$n =$ no. of states

Let

$$P_{\pi} = \begin{bmatrix} P_{\pi}(1,1) & \dots & P_{\pi}(1,n) \\ P_{\pi}(2,1) & \dots & P_{\pi}(2,n) \\ \vdots & \vdots & \vdots \\ P_{\pi}(n,1) & \dots & P_{\pi}(n,n) \end{bmatrix}_{n \times n}$$

Here

$$P_{\pi}(i,j) = \sum_a \pi(a|i) \sum_r P(j,r|i,a)$$

Let

$$R_{\pi} = \begin{bmatrix} R_{\pi}(1) \\ R_{\pi}(2) \\ \vdots \\ R_{\pi}(n) \end{bmatrix}$$

where $R_{\pi}(i) =$

$$\sum_a \pi(a|i) \sum_{r, r'} P(r'|i, a) r,$$

$i = 1, \dots, n$

Recall

$$V_{\pi}(s) = \sum_a \pi(a|s) \sum_{s', r} P(s', r|s, a) * [r + \gamma V_{\pi}(s')]$$

This equation can now be rewritten as

$$\checkmark V_{\pi} = R_{\pi} + \gamma P_{\pi} V_{\pi}$$

$$\Rightarrow (I - \gamma P_{\pi}) V_{\pi} = R_{\pi}$$

$$\text{where } I = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}_{n \times n}$$

$$\Rightarrow V_{\pi} = (I - \gamma P_{\pi})^{-1} R_{\pi}$$

- Inverse exists since $\gamma < 1$.

Iterative solution procedure for finding

V_{π} : [Iterative Policy Evaluation]

- Start with an initial estimate V_0 of the value function V_π
- Iterate over $k \geq 0$ to obtain the following update:

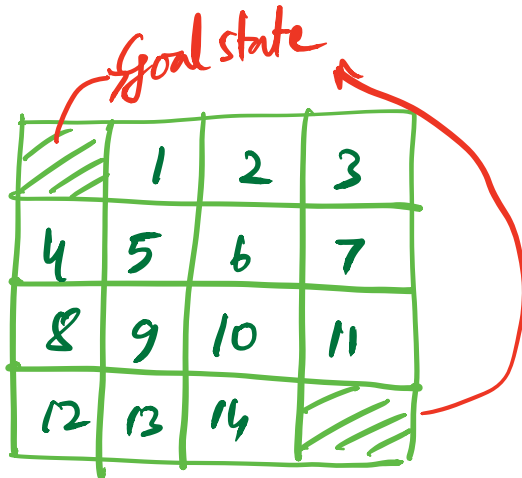
$$V_{k+1}(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r|s, a) * [r + \gamma V_k(s')], \quad \forall s \in S$$

It can be shown that as $k \rightarrow \infty$,

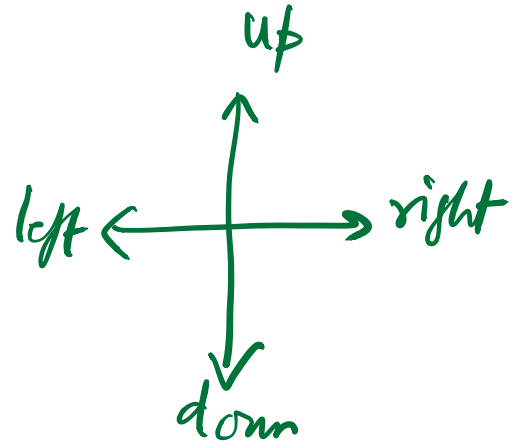
$$V_k(s) \rightarrow V_\pi(s) \quad \forall s$$

Example

Consider a 4×4 grid world



States



Actions

Nonterminal states:

$$S = \{1, 2, 3, \dots, 14\}$$

$$A = \{up, down, right, left\}$$

- Actions move state one step in suggested direction
- Actions that lead the agent out of grid leave state unchanged
- Reward = -1 on all transitions until termination
- $\gamma = 1$
- value of goal states = 0.
- policy : Equiprobable policy π

$$\begin{aligned}\pi(\text{up} | s) &= \pi(\text{down} | s) = \pi(\text{right} | s) \\ &= \pi(\text{left} | s) = 1/4\end{aligned}$$

We initialize all values to 0.

V_k for Random policy

$k=0$

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

Greedy policy w.r.t. V_k

///	↕	↕	↕
↕	↕	↕	↕
↕	↕	↕	↕
↕	↕	↕	///

$k=1$

0	-1	-1	-1
-1	-1	-1	-1
-1	-1	-1	-1
-1	-1	-1	0

///	←	↕	↕
↑	↕	↕	↕
↕	↕	↕	↓
↕	↕	→	///

$k=2$

0	-1.7	-2	-2
-1.7	-2	-2	-2
-2	-2	-2	-1.7
-2	-2	-1.7	0

///	←	←	↕
↑	↖	↕	↓
↑	↕	↘	↓
↕	→	→	///

$$U_2(1) = -1 + \frac{2}{4}(-1) + \frac{1}{4} * 0$$

$$= -1 - 0.75 = -1.75$$

$R=3$

0	-2.4	-2.9	-3
-2.4	-2.9	-3	-2.9
-2.9	-3	-2.9	-2.4
-3	-2.9	-2.4	0

	←	←	↖
↑	↖	↖	↓
↑	↔	↗	↓
↙	→	→	

$$U_3(1) = -1 + \frac{1}{4}[-1.7 - 2 - 2 + 0]$$

$$= -1 + \frac{1}{4}[-5.4]$$

$$= -1 - 1.4$$

$$= -2.4$$

$R=00$

0	-14	-20	-22
-14	-18	-20	-20
-20	-20	-18	-14
-22	-20	-14	0

7

11

///	←	←	↖
↑	↖	↖	↓
↑	↗	↗	↓
↘	→	→	///

Exercise 4.1

Find

(i) $q_{\pi}(11, \text{down})$, (ii) $q_{\pi}(7, \text{down})$

$$q_{\pi}(11, \text{down}) = -1 + 0 = -1$$

$$> v_{\pi}(11)$$

$$q_{\pi}(7, \text{down}) = -1 - 14$$

$$= -15 > v_{\pi}(7)$$

Exercise 4.2

In this example, suppose state 15 is added just below state 13 with

following actions:

///	1	2	3
4	5	6	7
8	9	10	11
12	13	14	///
	15		

(15) $\xrightarrow{\text{left}}$ (12)

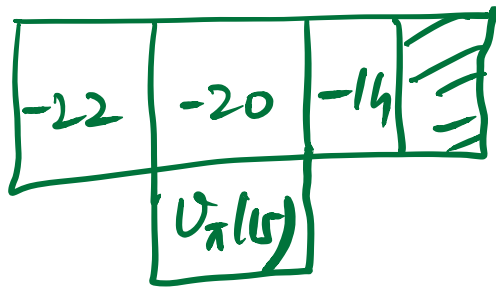
(15) $\xrightarrow{\text{right}}$ (14)

(15) $\xrightarrow{\text{up}}$ (12)

(15) $\xrightarrow{\text{down}}$ (15)

Assume transitions from original states are unchanged.

Find $V_{\pi}(15)$?



$$V_{\pi}(15) = -1 + \frac{1}{4} (-22 - 20 - 14 + V_{\pi}(15))$$

$$= -1 - 14 + \frac{1}{4} V_{\pi}(15)$$

$$\therefore \frac{3}{4} V_{\pi}(15) = -15$$

$$\therefore U_{\pi}(15) = \frac{-15 \times 4}{2} \\ = -20$$

What if only up & down actions are allowed in (15).

$$U_{\pi}(15) = -1 + \frac{1}{2} (-20 + U_{\pi}(15)) \\ = -1 - 10 + \frac{1}{2} U_{\pi}(15)$$

$$\Rightarrow \frac{1}{2} U_{\pi}(15) = -11$$

$$\checkmark U_{\pi}(15) = -22$$

Suppose action a in state s is
advantageous or could lead to a disadvantage.

We need to verify whether

$$q_{\pi}(s, a) \geq U_{\pi}(s)$$

- Let π & π' be two policies s.t.

$$q_{\pi}(s, \pi'(s)) \geq U_{\pi}(s) \quad \forall s \in \mathcal{S}$$

Claim: This implies $U_{\pi'}(s) \geq U_{\pi}(s)$.

Notice:

$$U_{\pi}(s) \leq q_{\pi}(s, \pi'(s))$$

$$= E \left[R_{t+1} + \gamma V_{\pi}(s_{t+1}) \mid s_t = s, A_t = \pi'(s) \right]$$

$$= E_{\pi'} \left[R_{t+1} + \gamma V_{\pi}(s_{t+1}) \mid s_t = s \right]$$

$$\leq E_{\pi'} \left[R_{t+1} + \gamma q_{\pi}(s_{t+1}, \pi'(s_{t+1})) \mid s_t = s \right]$$

$$= E_{\pi'} \left[R_{t+1} + \gamma E_{\pi'} \left[R_{t+2} + \gamma V_{\pi}(s_{t+2}) \mid s_{t+1}, A_{t+1} = \pi'(s_{t+1}) \right] \mid s_t = s \right]$$

$$= E_{\pi'} [R_{t+1} + \gamma R_{t+2} + \gamma^2 V_{\pi'}(s_{t+2}) | s_t = s]$$

$$\leq E_{\pi'} [R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+2} + \gamma^3 V_{\pi'}(s_{t+3}) \\ \vdots | s_t = s]$$

$$\leq E_{\pi'} [R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | s_t = s] \\ = V_{\pi'}(s)$$

Thus, if policy π' is s.t.

$$q_{\pi}(s, \pi'(s)) \geq V_{\pi}(s),$$

then $V_{\pi'}(s) \geq V_{\pi}(s) \quad \forall s \in S.$

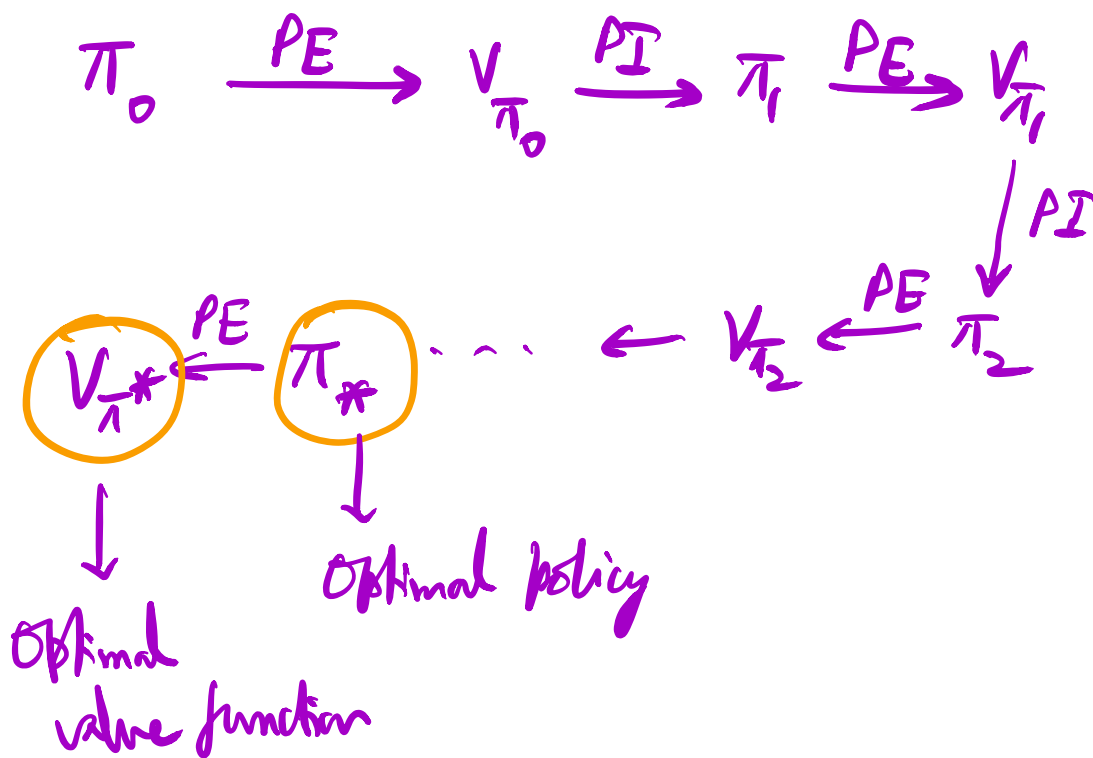
Suppose we obtain a greedy policy π' as

$$\begin{aligned} \pi'(s) &= \underset{a}{\operatorname{argmax}} q_{\pi}(s, a) \\ &= \underset{a}{\operatorname{argmax}} E \left[R_{t+1} + \gamma V_{\pi}(s_{t+1}) \mid \right. \\ &\quad \left. s_t = s, A_t = a \right] \end{aligned}$$

$$= \underset{a}{\operatorname{argmax}} \sum_{s', r} p(s', r | s, a) [r + \gamma V_{\pi}(s')]$$

Policy Improvement Step

Policy Iteration Procedure



Note: No. of policies is finite

Since no. of states & actions are finite.

Policy Iteration Converges in a finite no. of steps.

Policy Iteration (using Iterative PE)

1. Initialization

$V(s) \in \mathbb{R}$, $\pi(s)$ arbitrary

2. Policy Evaluation

Loop:

$\Delta \leftarrow 0$

Loop for each $s \in S$:

$v \leftarrow V(s)$

$$V(s) \leftarrow \sum_{s', r} P(s', r | s, \pi(s)) [r + \gamma V(s')]$$

$$\Delta \leftarrow \max(\Delta, |V - V(s)|)$$

until $\Delta < \theta$ (a small positive
no. determining
accuracy of estimation)

3. Policy Improvement

For each $s \in S$:

$$\text{old-action} \leftarrow \pi(s)$$

$$\pi(s) \leftarrow \underset{a}{\operatorname{argmax}} \sum_{s', r} P(s', r | s, a) [r + \gamma V(s)]$$

- If $\text{old-action} = \pi(s) \neq s$,

then stop & return $V \approx V^*$
 $\pi \approx \pi^*$

- If old-action $\neq \pi(s)$,
go back to step 2 & perform
PE for policy π .

If $V_{\pi}(s) = V_{old}(s) \forall s$,
then $V_{\pi} = V_*$

& both π & π_{old} are
optimal

Value Iteration Procedure

Recall the Bellman Equation for Optimality:

$$V_*(s) = \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma V_*(s')] \quad \forall s \in S$$

Value Iteration procedure

— starts with an initial estimate

V_0 of V_*

— Iterate

$$V_{k+1}(s) = \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma V_k(s')]$$

$$\forall s \in S$$

It turns out that as $k \rightarrow \infty$,

$$V_k(s) \rightarrow V_*(s) \quad \forall s \in S.$$

Value Iteration for Estimating $\pi \approx \pi^*$

- Algorithm parameter:

a small threshold $\theta > 0$

determining the estimation accuracy.

- Initialize $V(s) \quad \forall s \in S^+$ arbitrarily
except that $V(\text{terminal}) = 0$.

- Loop :

$$\Delta \leftarrow 0$$

Loop for each $s \in S$:

$$v \leftarrow V(s)$$

$$V(s) \leftarrow \max_a \sum_{s', r} P(s', r | s, a) * [r + \gamma V(s')]$$

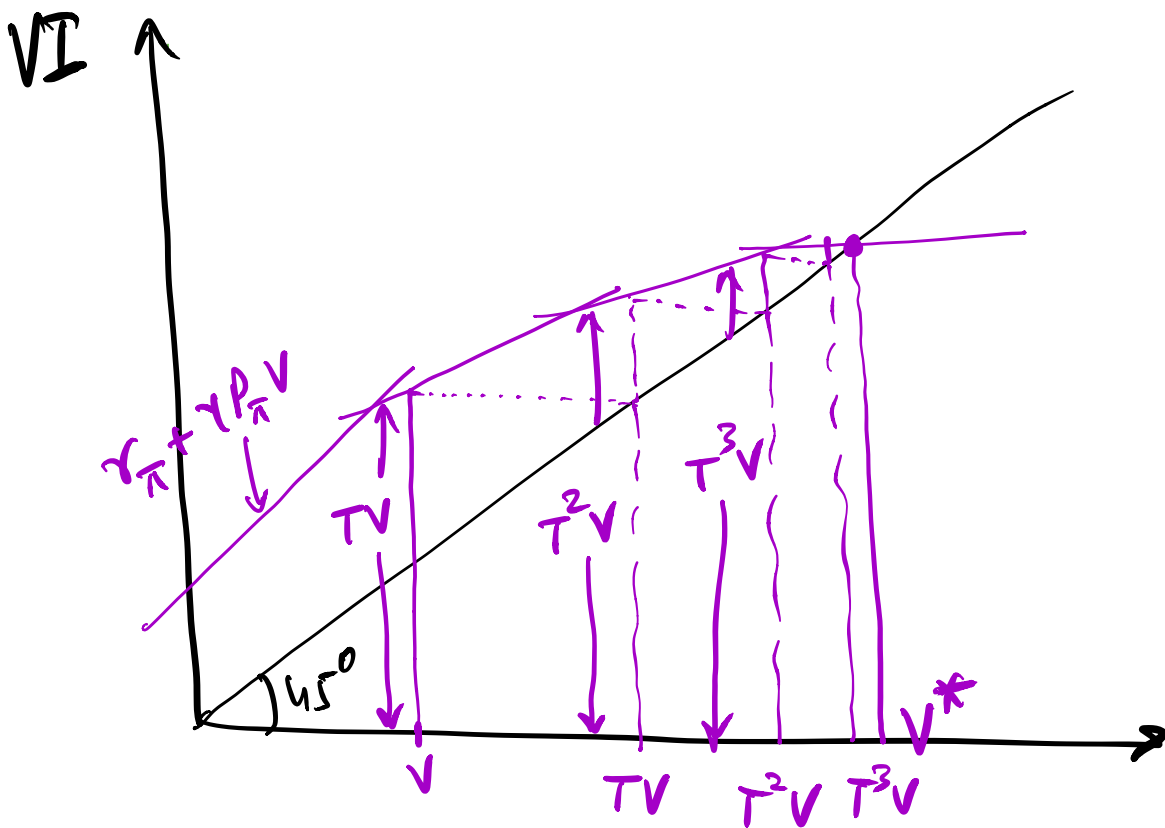
$$\Delta \leftarrow \max(\Delta, |v - V(s)|)$$

Until $\Delta < \theta$

- Output a deterministic policy $\pi \approx \pi_*$
s.t.

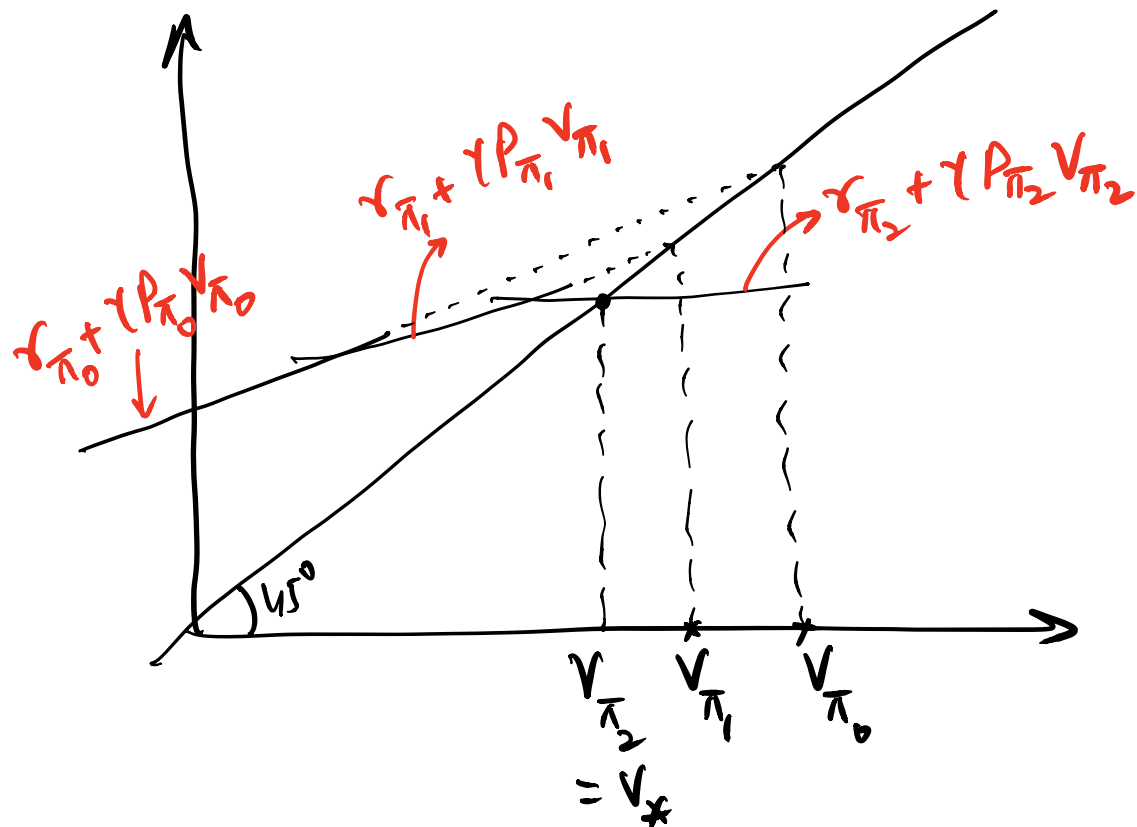
$$\bar{\pi}(s) = \operatorname{afmax}_a \sum_{s', r} P(s', r | s, a) [\bar{r} + \gamma V(s')] \\ \forall s \in S.$$

Geometric Interpretation of VI & PI



AI

$$TV = \max_{\pi} [R_{\pi} + \gamma P_{\pi} V]$$

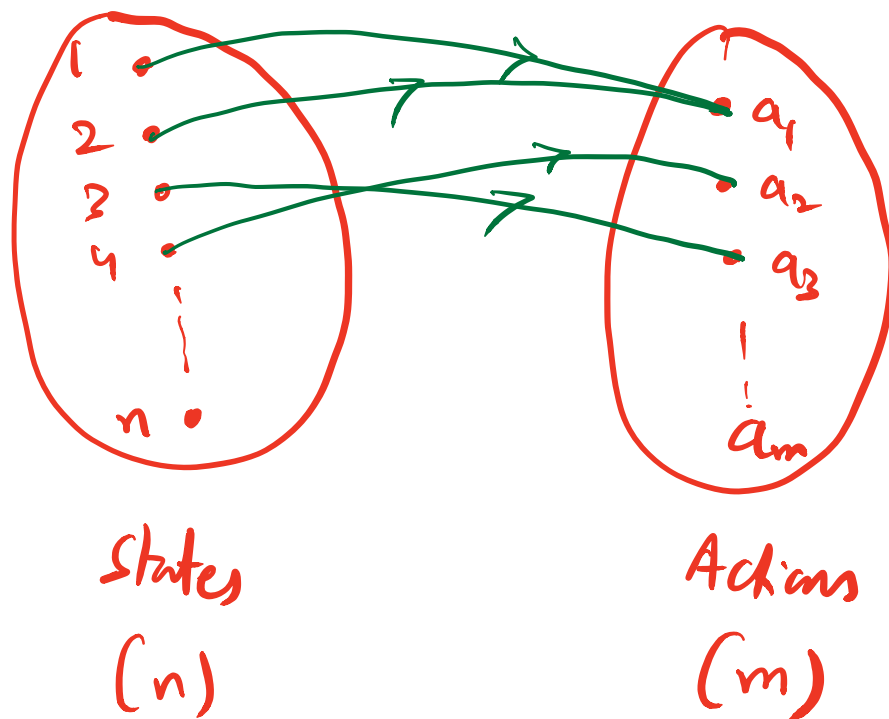


$$V_{\pi_k} \rightarrow V_* \quad \text{and} \quad \pi_k \rightarrow \pi_*$$

as $k \rightarrow \infty$.

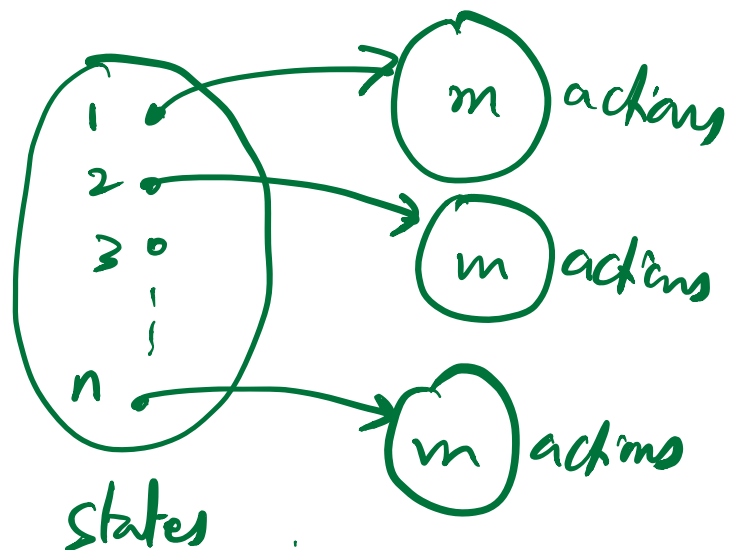
Note: Value Iteration Can take a
Large no. of iterations to Converge but
Policy Iteration Converges in a small (finite)
no. of iterations

Policies



A deterministic policy is a map from states to actions

Q) How many functions exist between state & action spaces



Each state has a choice of m actions

∴ Total no. of functions

$$= \underbrace{m \times m \times m \times \dots \times m}_{n \text{ times}}$$

$$= m^n$$

$$\Rightarrow \text{Total no. of deterministic policies}$$

$$= m^n < \infty.$$

In PI, since we strictly improve in every step of PI, the procedure converges in a finite no. of iterations to the optimal policy.

Next class onwards

→ We shall consider situations
where system model, $p(s', r | s, a)$,
is not known but data is available
(real or simulated data)

→ Data is in the form of tuples

$(s_0, A_0, R_1, s_1),$

$(s_1, A_1, R_2, s_2),$

$\vdots$

Goal of Chapter 5:

Find optimal policies & value functions only using data samples that are available.